Research Interests: Design, simulation, and fabrication of autonomous vehicles, surface vessels, and hydro-mechanical systems, including hydraulically actuated mechanisms. Areas of specific interest include hydrodynamics, AUV dynamics, and computational fluid dynamics.

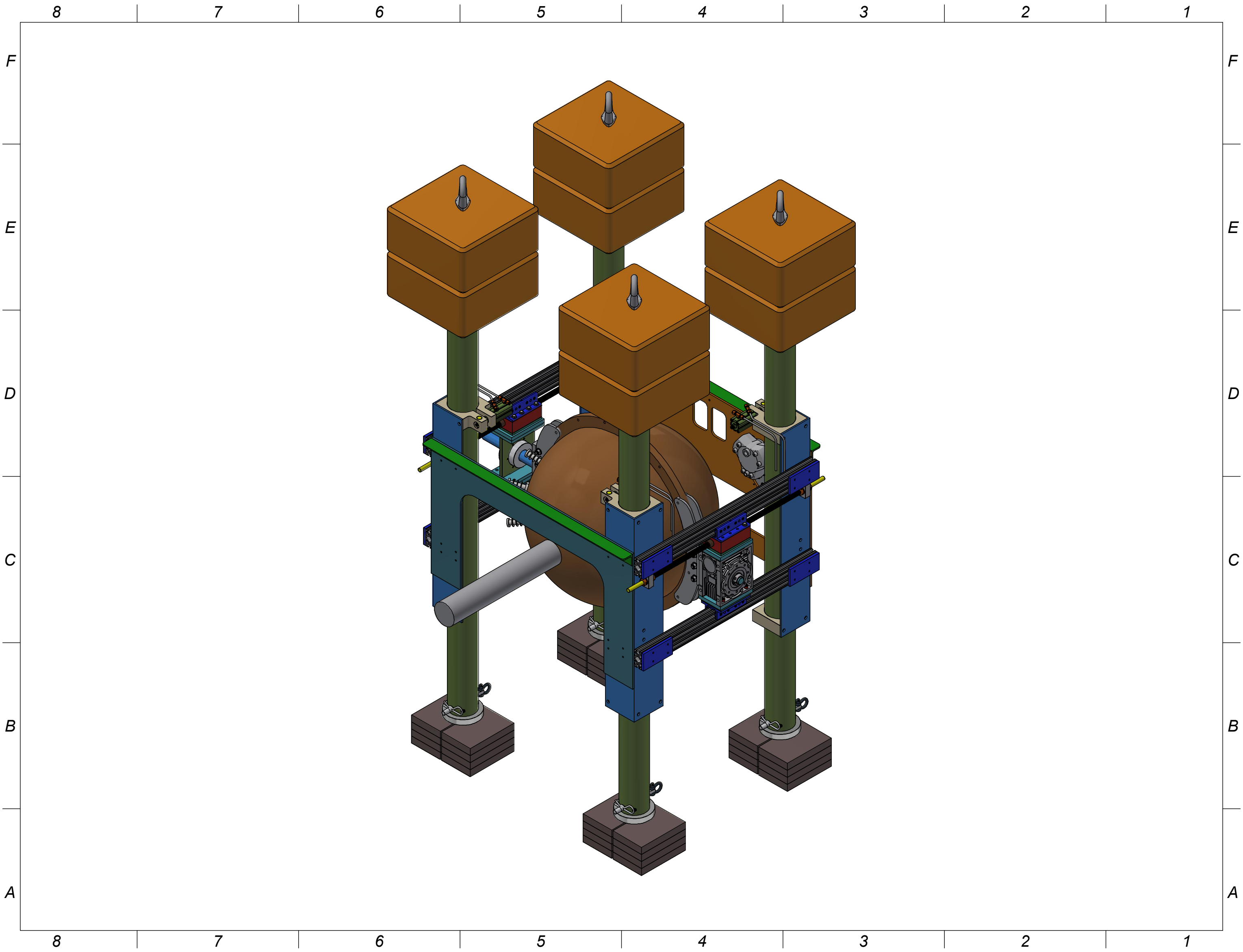
Main projects I have worked on since joining WHOI (2023–2026):

1. Ocean Reference Stations (Stratus and WHOTS) – 2023, 2024, 2025, 2026
2. Wind Forecast Improvement Project (WFIP-3) – 2023, 2024
3. ASTRAL/EKAMSAT – 2024, 2025
4. Ocean Margins Initiative (Ghana) – 2025, 2026
5. SAFARI – 2025
6. LIBS (Laser Induced Breakdown Spectroscopy) – 2025, 2026
7. Endurance Energy – 2026
8. Spar Buoy – 2026
9. microAlto profiling float (CFD modeling) – 2026

Roles & Contributions across Projects (see the pictures below):

- Took over a time-critical marine engineering project, advancing the mechanical design of a subsea elevator system for deep-sea operations with a ROV (2600 m water depth).
- Performed hydrodynamic analysis to evaluate surface and subsurface buoy/mooring response, knockdown, and load distribution.
- Ran computational fluid dynamics (CFD) for USVs, AUVs, and surface buoys to test performance, optimize designs for stability and efficiency, and predict behavior under various environmental conditions.
- Supported deck operations, deployment, recovery, and real-time troubleshooting during research cruises.
- Participated in the piloting of USV platforms.
- Designed mechanical components and instrument mounting assemblies for surface and subsurface buoys/moorings and Wave Gliders.

- Integrated meteorological and oceanographic sensors into field-ready buoy and USV systems.
- Coordinated with field teams to align mechanical design with deployment and operational constraints.
- Fabricated mechanical components in-house, including cutting parts on a waterjet machine.
- Utilized 3D printing for both prototyping and final deployed components, selecting materials suitable for marine environments (e.g., UV and saltwater exposure).
- Collaborated with machine and fabrication shops to produce custom mechanical components.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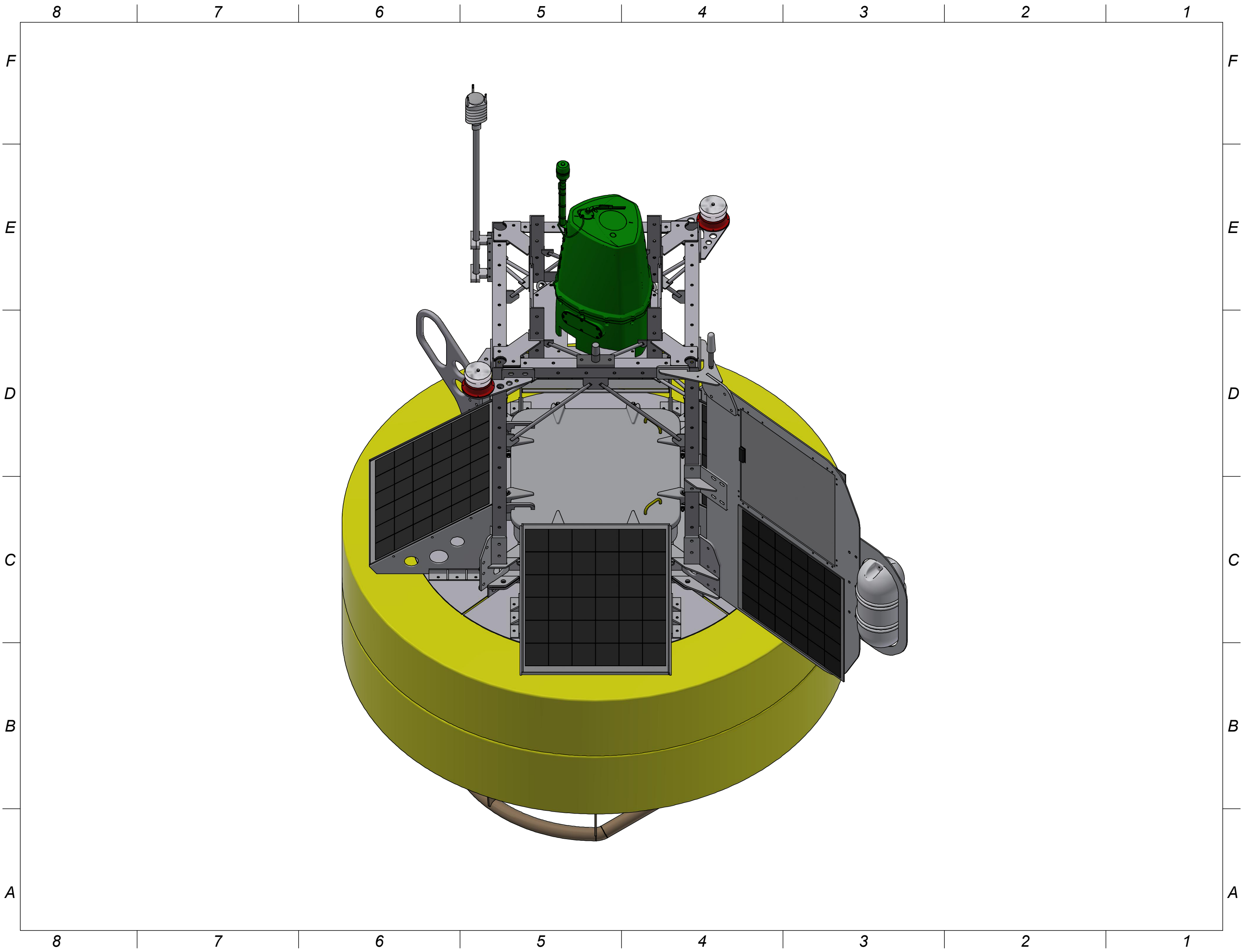
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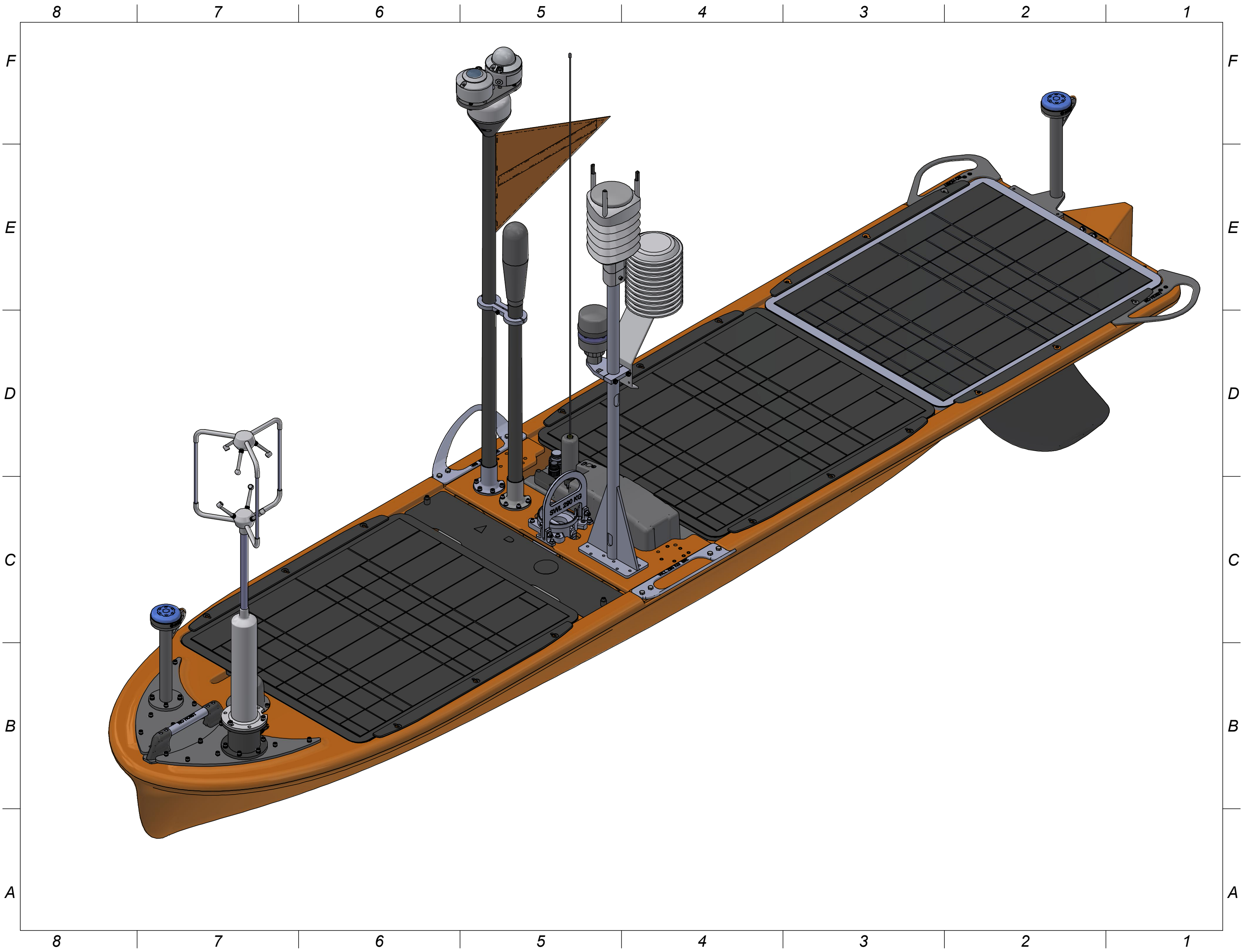
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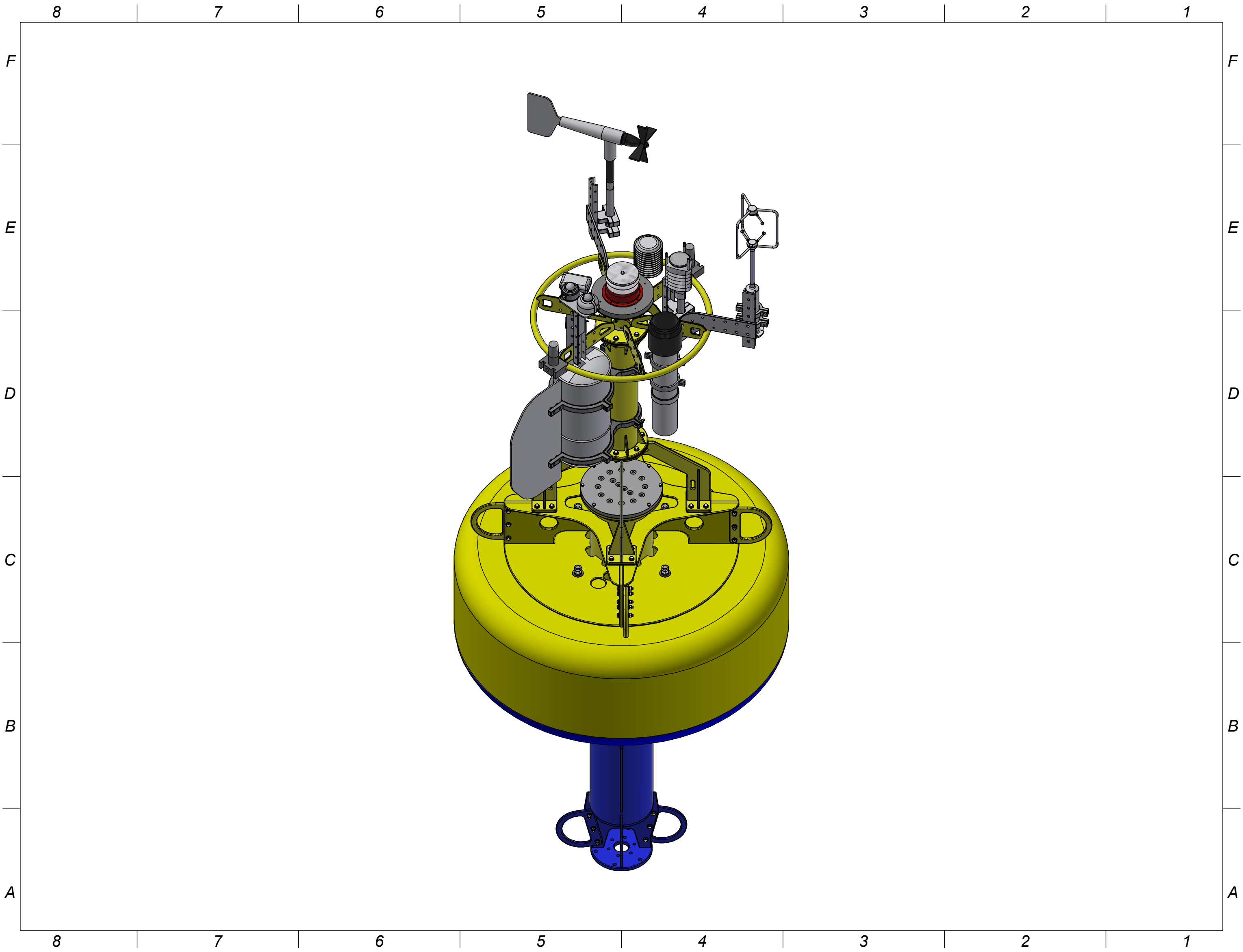
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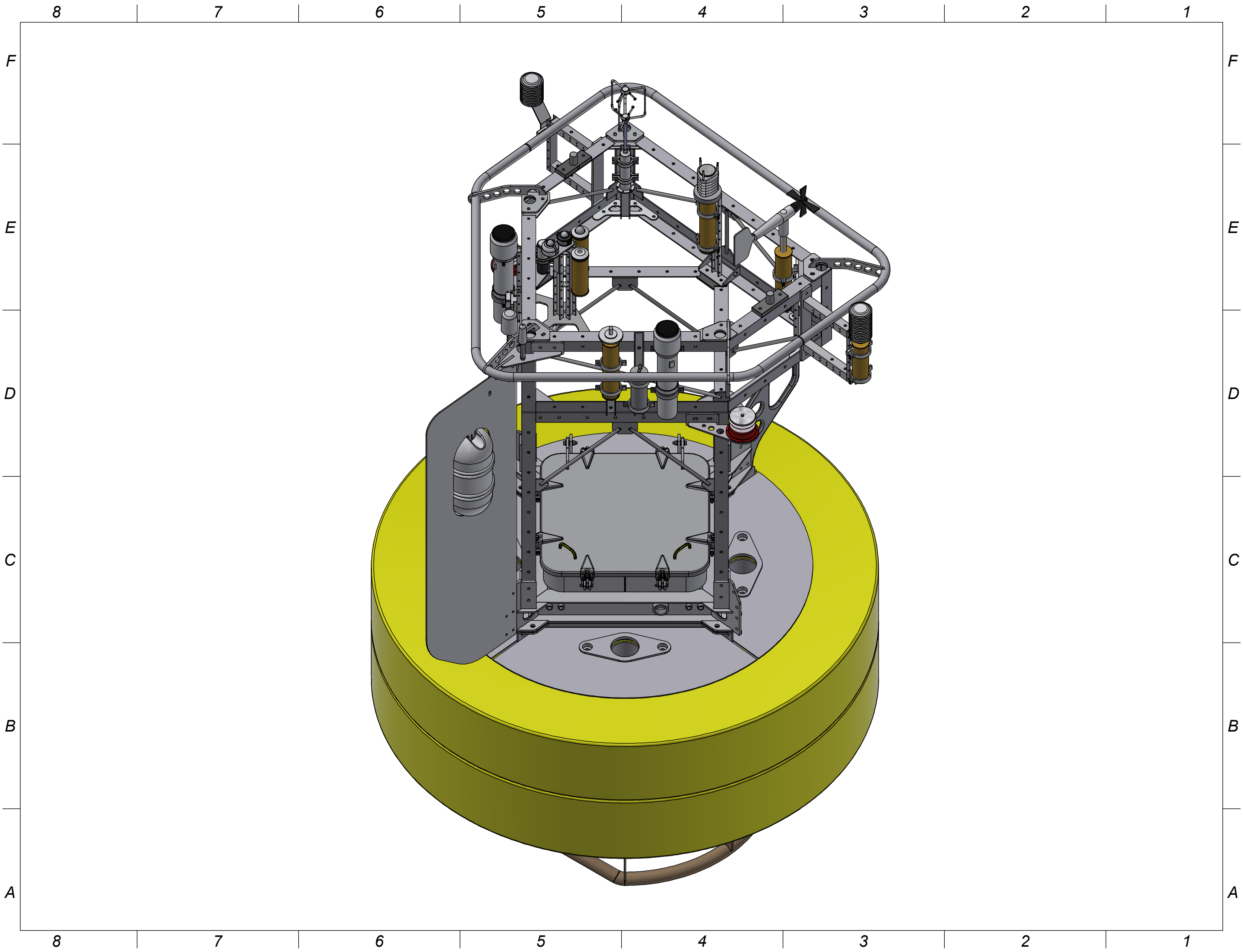
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Additional Technical Work (M.S. Thesis):

Conducted CFD analysis on WHOI stretch-hose mooring fitted with helical grooves to optimize groove geometry for reduced drag and mitigation of vortex-induced vibration (VIV).